

Grade Eight Science - Chemistry Lab Information Sheet (CRITERION B)

Title of the Experiment:

Will the temperature of hydrogen peroxide change if a beef liver catalyst is added?

What you have to do:

- Written below is the Criterion B section of the lab - Criterion B is given to you but you have to complete Criterion C
- Your job is complete the experiment by following the method given below and collect the data
- Then you must complete the Criterion C worksheet for this experiment

Background Information:

- ☐ **Describe** the validity of the research question be tested by scientific investigation
- ☐ **Explain** the reasoning behind the hypothesis using correct scientific reasoning
- ☐ In-text citations for sources used must be present in this section

Validity of the Research Question:

The purpose of this experiment is to see if hydrogen peroxide changes in temperature if a catalyst of beef liver is added, and to analyze this temperature data to see if a chemical change has occurred or not.

The research question is valid because understanding temperature changes teaches students to distinguish between physical and chemical changes; a foundation in every Chemistry course. A physical change keeps the starting material and does not produce anything new, while a chemical change uses up the starting material and produces new substances. This experiment connects classroom science to daily life. For example, each day we observe temperature changes by using items such as hot packs and cold packs, and we observe the result of combustion reactions in car engines and gas barbeques ("Chemical and Physical Change")

Reasoning behind the Hypothesis:

A catalyst speeds up the rate of a chemical change. Beef liver contains a chemical that catalyzes the breakdown of hydrogen peroxide into hydrogen gas and water. The beef liver will break down the hydrogen peroxide and, therefore, may absorb or produce heat over the course of the reaction resulting in a temperature change ("What does beef liver do as a catalyst?").

Research Question:

- ☐ Must be based on the **description** given in the **Background Information**
- ☐ Must be in the form of a question

If a catalyst of beef liver is added to hydrogen peroxide will the temperature change

indicating that a chemical change has happened?

Hypothesis:

- ☐ Must be based on the reasoning given in the **Background Information**
- ☐ **State** the hypothesis in an "if, then, because," format
- ☐ Must **state** all variables used, followed by **one** chosen variable

If a catalyst of beef liver is added to hydrogen peroxide then the temperature of the hydrogen peroxide will drop, indicating that a chemical change has happened, because the beef liver will absorb heat from the hydrogen peroxide causing it to become colder

Variables:

- ☐ **Identify** the dependent variable and the independent variable
- ☐ **Describe** how the data for each variable will be collected

Variable	The variable is:	Method of Measurement
Dependent	The temperature of the hydrogen peroxide	• The temperature of the hydrogen peroxide is measured in degrees Celsius using a glass immersion thermometer • The temperature will be read from the top of the red line once the line stops moving • The temperature will be recorded in a data table • The temperature will be taken before and after the addition of beef liver • The temperature will be taken three times and averaged
Independent	The presence or absence of the beef liver	• The temperature of the hydrogen peroxide will be taken using the above described method before and after beef liver is added • All temperatures will be taken three times and averaged

Control Variables:

- ☐ **Identify** each control variable (**add more rows to the table if needed; not all rows need to be used**)
- ☐ **Describe** the method used to allow each variable to be controlled

The Control Variable is:	Method of Control
The method of measurement of temperature for the hydrogen peroxide	• The same thermometer was used for each trial • The thermometer was read from the top of the red line once it stopped moving
Volume of hydrogen peroxide used for each trial	• The same volume of hydrogen peroxide was used for each trial (50 mL) • The hydrogen peroxide was measured using the same graduated cylinder each time
Type of liver being used	• Beef liver from the same package is used in each trial
Reading of the graduated cylinder	• All graduated cylinder readings are read from the bottom of the meniscus

Materials:

- ☐ Should be in a list (not a paragraph); you can bullet the list
- ☐ Should be as specific as possible (ie. don't just say "beakers," say Ten 250 mL beakers, for example)

- Hydrogen Peroxide (150 mL)
- Immersion Thermometer (1)
- 250 mL Beakers (3)
- Beef liver (3 pieces)
- Painters tape (3 strips)
- 100 mL Graduated cylinder (1)
- Tap Water (as much as needed)
- Paper towel (as much as needed)
- Weighing boat

Method:

- ☐ Should be written in the past tense
- ☐ Should be numbered - do not use bullet points
- ☐ Should be clear and concise - **repeat steps** should be used wherever possible
- ☐ Should be logical, safe, and complete

1. Three 250 mL beakers were labelled "Trial 1, Trial 2, and Trial 3," using painters tape and a pen
2. 50 mL of hydrogen peroxide was poured from its container into a graduated cylinder
3. The 50 mL of hydrogen peroxide was poured from the graduated cylinder into a 250 mL beaker
4. Steps 2 and 3 were repeated 2 more times using the Trial 2 and 3 beakers
5. An immersion thermometer was used to measure the temperature of the hydrogen peroxide in the Trial 1 beaker.
6. The temperature was read from the top of the red line once the line stopped moving
7. The temperature was recorded in the data table
8. A piece of beef liver was dropped into the Trial 1 beaker
9. Steps 6 and 7 were repeated
10. The thermometer was removed from the Trial 1 beaker, rinsed off with water from the tap and dried off with a paper towel.
11. Steps 5 to 10 were repeated for the Trial 2 beaker
12. Steps 5 to 10 were repeated for the Trail 3 beaker

MLA Works Cited:

"Chemical and Physical Change Examples." *FCT EMIS*. 27 April 2026.
https://fctemis.org/notes/9942_Chemical%20And%20Physical%20Change%20Example1.pdf

"What does beef liver do as a catalyst?" prompt Google Gemini. 27 April, 2026.
gemini. google.com